Biological Organization

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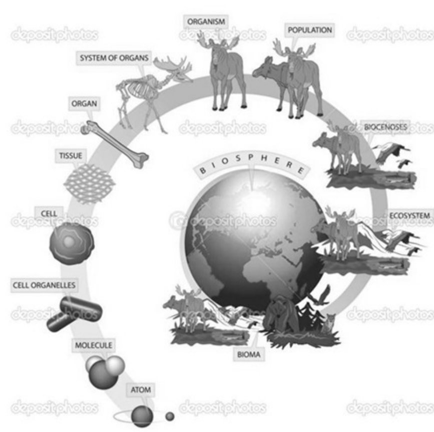
Levels of biological organization

It is the hierarchy of complex biological structures and systems that define life using a reductionist approach



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Levels of biological organization

Biosphere
⬇
Ecosystem
⬇
Community
⬇
Population
⬇
Organism
⬇
Organ System
⬇
Organ
⬇
Tissue
⬇
Cell
⬇
Organelle
⬇
Molecule
⬇
Atom

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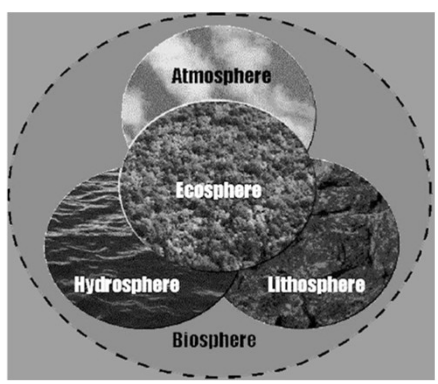
Biosphere

- All regions of earth's surface, water and atmosphere.
- It is the worldwide sum of all ecosystems. It can also be termed as zone of life on Earth and largely self-regulating



Ecosystem

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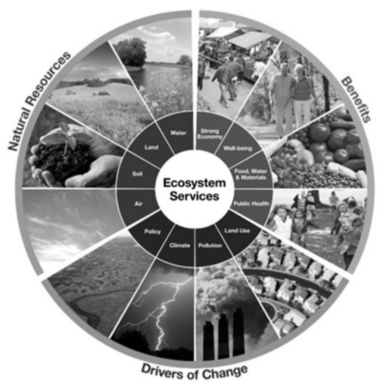
Ecosystem

- An ecosystem is a community made up of <u>living</u> organisms and <u>non-living</u> components such as air, water, minerals and soil or it could be temperature, precipitation and surrounding geography.
    - ❖ Desert
    - ❖ Pond
    - ❖ River
    - ❖ Lake



Community

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Community

- A community is a small or large social unit that has something in common such as norms, religion, values, or identity.
- Interactions of different populations of species living in one ecosystem.
- e.g., a forest of trees and undergrowth plants, inhabited by animals and rooted in soil containing bacteria and fungi, constitutes a biological community.

Population

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Population

- A group of members of the same species living in a given area.
- It is important because reproduction and genetic changes occur in members of a species.

Organism

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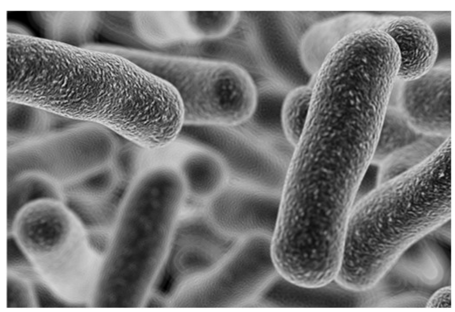
Organism

- The basic living system of any given population such as animals, plants, microorganisms.
- It can be a single-celled life form.



Organ system

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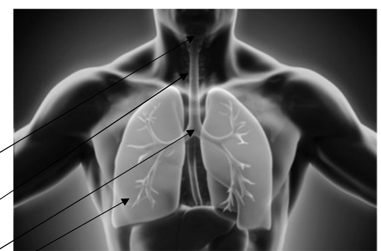
Organ system

- A group of organs that together work to fulfill a larger role for a multi-cellular organism.
    - Skeletal system
    - Circulatory system
    - Respiratory system
    - Nervous system



Larynx
Trachea
Bronchus
Lung

Organ

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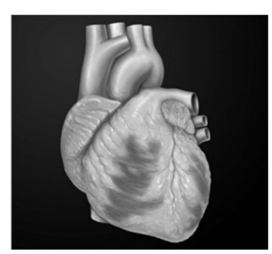
Organ

- An organ is a collection of tissues that carry out a collective function.
    - Lung
    - Heart
    - Leg
    - Hand
    - Stomach



Tissue

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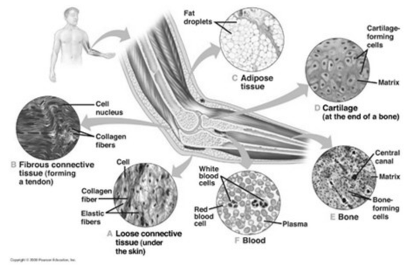
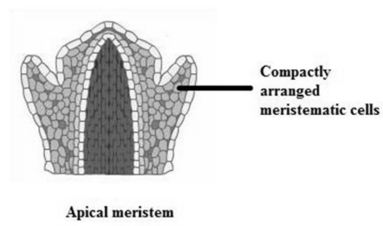
Tissue

- Tissues are collection of cells and other materials that work together to carry out a very specific function for a multi-cellular organism.
- Animals – Connective, muscle, epithelial
- Plants – Meristematic, Parenchyma, Collenchyma





Cell

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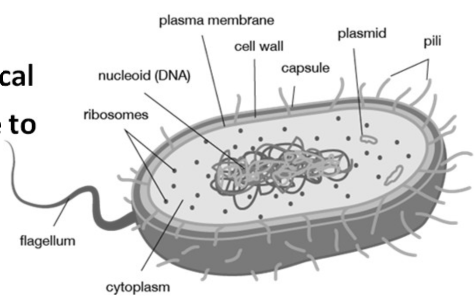
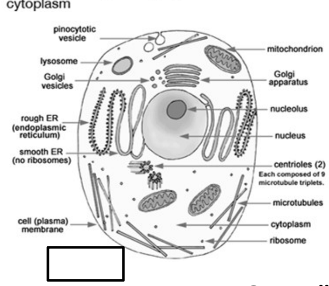
Cell

- Smallest level of biological organization that is able to live independently.
  - Prokaryotic Cells
  - Eukaryotic Cells





Organelle

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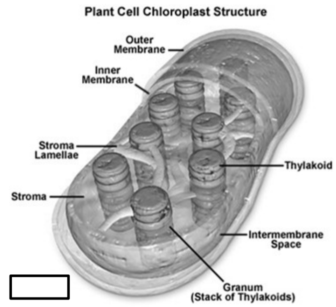
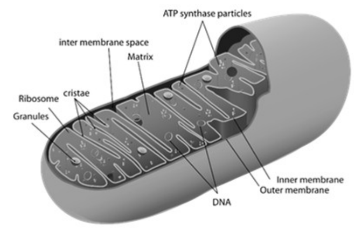
Organelle

- They are the membrane bound compartments found within eukaryotic cells.
- Each organelle carry out a specific function.
  - Mitochondria
  - Chloroplast
  - Golgi bodies
  - Nucleus





Molecule

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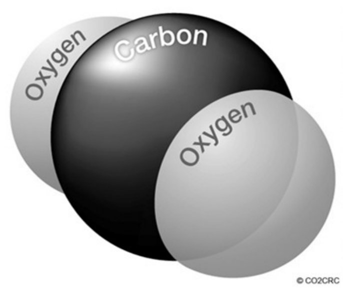
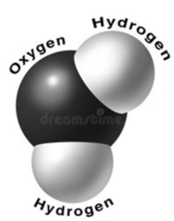
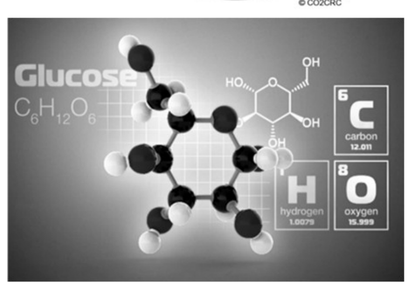
Molecule

- Two or more atoms join together to form a molecule.
  - CO_2
  - Proteins
  - Carbohydrates
  - Lipids







Atom

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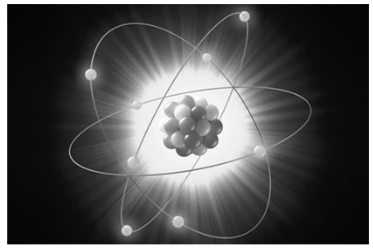
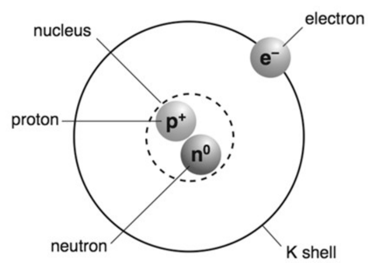
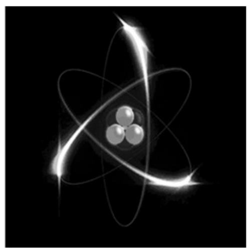
Atom

- It is the smallest identifiable form of any element.
- We also have subatomic particles
  - Protons
  - Neutrons
  - Electrons







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⬇	⬇
Ecosystem	Tissue
⬇	⬇
Community	Cell
⬇	⬇
Population	Organelle
⬇	⬇
Organism	Molecule
⬇	⬇
Organ System	Atom
⬇	

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⬇	

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THANK YOU

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